YASKAWA

MLX300
SOFTWARE OPTIONS
USERS MANUAL

FOR: CONVEYOR TRACKING

Upon receipt of the product and prior to initial operation, read these instructions thoroughly, and retain for future reference.

Have the following information available when contacting the YASKAWA Representative:

- System
- Primary Application
- Software Version *(Located on Programming Pendant by selecting: {Main Menu} - {System Info} - {Version})*
- Warranty ID *(Located on Robot Controller)*
- Robot Serial Number *(Located on Manipulator data plate)*
- Robot Sales Order Number *(Located on Robot controller data plate)*

The Americas YASKAWA Representative

- 24-hour Telephone Number: **(937) 847-3200**
  Use for urgent or emergency needs for technical support, service and/or replacement parts
- Routine Technical Inquiries: *techsupport@motoman.com*
  Allow up to 36 hours for response

MANUAL NO.
188068-1CD ②

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www.motoman.com

Safety

For Your Safety

Robots generally have requirements which are different from other manufacturing equipment, such as larger working areas, high-speed operation, rapid arm movements, etc., which can pose safety hazards.

Read and understand the instruction manuals and related documents, and observe all precautions in order to avoid the risk of injury to personnel and damage to equipment.

Carelessness contributes to serious accidents in the work area.

It is the user's responsibility to ensure that all local, state, and national codes, regulations, rules, or laws relating to safety and safe operating conditions are met and followed.

DANGER

- Teaching, operations, and maintenance of the Robot must conform to:
  - Industrial Safety and Health Law
  - Order for Enforcement of the Industrial Safety and Health Law
  - Industrial Safety and Health Regulations
  - Technical Standards for Electrical Facilities

  Other related laws and regulations are:
  - Occupational Safety and Health Act in USA
  - Factory Act (Gewerbeordnung) in Germany
  - Health and Safety at Work, etc. Act in UK
  - EC Machinery Directive 2006/42/EC
- Prepare:
  - SAFETY WORK REGULATIONS
    based on concrete policies for safety management complying with related laws and regulations.
- Observe:
  - JIS B 8433-1: 2015 "Robots for industrial environments-Safety requirements" (ISO 10218-1: 2011) for safe operation of the robot. (JIS B 8433 is for Japan only)
- Reinforce:
  - SAFETY MANAGEMENT SYSTEM
    by designating authorized operators and safety managers for the Robot, as well as giving continuing safety education and training.
- Teaching, operation, and maintenance of the Robot are specified as "Hazardous Operations" in the Industrial Safety and Health Act (for Japan only).
  Personnel engaged in these operations must receive special training offered by YASKAWA.

DANGER

- Personnel engaged in operation, maintenance, or management of the Robot must receive required training before using the Robot.

For more information on training, contact a YASKAWA representative.

- Make sure to have and follow all manuals, read them thoroughly and understand the contents of them.

Confirm that you have all required manuals. If any of the manuals are missing, contact a YASKAWA representative.

- Read and understand these instructions throughly before installing, operating, or maintaining the Robot.

Any matter not described in this manual must be regarded as “prohibited” or “improper”.

We suggest that you obtain and review a copy of the ANSI/RIA National Safety Standard for Industrial Robots and Robot Systems (ANSI/RIA R15.06-2012). You can obtain this document from the Robotic Industries Association (RIA) at the following address:

Robotic Industries Association
900 Victors Way
P.O. Box 3724
Ann Arbor, Michigan 48106
TEL: (734) 994-6088
FAX: (734) 994-3338
www.roboticsonline.com

Ultimately, well-trained personnel are the best safeguard against accidents and damage that can result from improper operation of the equipment. The customer is responsible for providing adequately trained personnel to operate, program, and maintain the equipment.

We recommend approved YASKAWA training courses for all personnel involved with the operation, programming, or repair of the equipment.

This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications.

WARNING

- Safe operation of this equipment is the users responsibility.
  - The conditions under which the equipment will be operated safely should be reviewed by the user. The user must be aware of the various national codes, ANSI/RIA R15.06-2012 safety standards, and other local codes that may pertain to the installation and use of this equipment.

Not following all national codes, safety standards and local codes can result in death or serious injury

- Additional safety measures for personnel and equipment may be required depending on system installation, operation, and/or location.

Not providing additional safety measures as required can result in death or serious injury.

- Check all safety equipment frequently for proper operation.
  - Repair or replace any non-functioning safety equipment immediately.

If safety equipment does not operate properly death or serious injury can result.

CAUTION

- Only trained personnel familiar with the operation, manuals, electrical design, and interconnections of this equipment should program, or maintain the system.

Any personnel involved with the operation of the equipment must understand potential dangers of operation.

NOTICE

- The drawings and photos in this manual are examples. Differences may exist between them and the delivered product.
- YASKAWA may modify this model without notice due to product improvements, modifications, or changes in specifications. If such modification is made, the manual number will also be revised.
- Some operations require standard passwords and while others require special passwords.
- If a manual is damaged or lost, contact a YASKAWA representative to order a new copy. Make sure to tell a YASKAWA representative the Manual Number listed on the front cover.

Notes for Safe Operation

Read this manual carefully before installing, operating, maintaining, or inspecting the system.

In this instruction, Safe Operations are classified as “DANGER”, “WARNING”, “CAUTION” or “NOTICE”.

DANGER	Indicates an imminently hazardous situation which, if not avoided, **WILL result in death or serious injury**.
WARNING	Indicates a potentially hazardous situation which, if not avoided, **MAY result in death or serious injury**.
CAUTION	Indicates a hazardous situation, which if not avoided, **MAY result in minor to moderate injury**.
CAUTION	Indicate a situation which if not avoided **may result in equipment damage.**
NOTICE	Indicates **practices not related to personal injury**.

NOTICE
To ensure safe and efficient operation at all times, be sure to follow all instructions, even if not designated as “DANGER”, “WARNING” or “CAUTION”.

Installation and Wiring Safety

Review the Robot and Controller Instructions for details on installation and wiring.

In planning installation, adapt an easy to observe arrangement to ensure safety. Take safety into consideration when planning the installation.

DANGER

- Any person who programs, teaches, operates, maintains or repairs the included system MUST be trained and demonstrates competence to safely perform assigned tasks.

Failure to observe these safety precautions may result in death or serious injury from unexpected movements.

Ensure Safety

DANGER

- Turn OFF servo power before operating.
  - Press the EMERGENCY STOP button to turn off SERVO POWER. When servo power is OFF, the SERVO ON LED on the Robot Programming Pendant is OFF.
  - Disconnect the safety plug of the safety fence. (When in PLAY mode or in REMOTE mode)

If the EMERGENCY STOP button(s) do not work correctly, death or serious injury may result. Do not use if the EMERGENCY STOP button does not perform correctly.

Fig. : EMERGENCY STOP Button

- Clear the cell of all items which could interfere with the operation before releasing the EMERGENCY STOP button.

Death or serious injury may result from unintentional or unexpected motion.

Fig. : Release of EMERGENCY STOP Button

- Observe the following precautions when performing a teaching operation within the robot's operating range:
  - Be sure to perform lockout by putting a lockout device on the safety fence when going into the area enclosed by the safety fence.
    - In addition, the operator of the teaching operation must display the sign that the operation is being performed so that no other person closes the safety fence.
  - View the manipulator from the front whenever possible.
  - Always follow the predetermined operating procedure.
  - Always keep in mind emergency response measures against the robot’s unexpected movement toward a person.
  - Ensure a safe place to retreat in case of emergency.

Failure to observe these instructions may cause improper or unintended movement of the robot, which may result in personal injury.

- Make sure no person is in the operating range before:
  - Turning ON power to the Robot Controller
  - Moving the Robot with the Robot Programming Pendant
  - Running the system in the TEACH mode
  - Performing automatic operations

Death or serious injury may result if a person enters the operating range during operation. Immediately press an EMERGENCY STOP button whenever there is a problem.

DANGER

- When the power supplies of the Robot and Robot Controller are turned ON at start-up, be sure to confirm the following:
  - Safety protection devices such as the E-STOP circuit, door interlocks, etc. operate normally.
  - Each axis operates normally in TEACH mode.
  - Robot operates normally at the speed limit or less in the TEACH mode.(Speed limit: 250 mm/s at the TCP or the flange)
  - The teaching function and the playback function operate normally.

DANGER

YRC1000micro/MLX300

- In the case of not using a Robot Programming Pendant, make sure to supply the EMERGENCY STOP button.
  - Connect the external EMERGENCY STOP button to the 4-14 pin and 5-15 pin of the Safety connector (Safety).
- Make sure EMERGENCY STOP button turns off servo power.
- Upon shipment of the robot controller, the EMERGENCY STOP signal is connected by a jumper cable using the safety signal short circuit connector.
  - To use the signal, make sure to supply a new connector, and then input it.

If the EMERGENCY STOP signal does not function, can result in death, serious injury or equipment damage.

DANGER

- When the power supplies of the Robot and Controller are turned ON at start-up, be sure to confirm the following:
  - Safety protection devices such as the E-STOP circuit, door interlocks, etc. operate normally.
  - Each axis operates normally in TEACH mode.
  - Robot operates normally at the speed limit or less in the TEACH mode.(Speed limit: 250 mm/s at the TCP or the flange)
  - The teaching function and the playback function operate normally.
- The Robot may stop movements while waiting for a condition to be satisfied during operation.

Once meeting the condition, the Robot starts movement causing a danger that will cause death or severe injury.

- Make sure to clearly indicate when the Robot is in operation:
  - Use a pilot lamp and/or an audible alert or
  - The Robot stops operation if the operator comes close.
- Install a safety fence around the Robot to prevent any accidental contact with the Robot when power is applied.
  - Display a warning sign stating “Off-Limits During Operation” at the entrance of the safety fence.
  - The gate of the safety fence must be equipped with a safety interlock to turn the servo power OFF when the gate opens.
  - Make sure interlocks operate properly before use.
- For areas not enclosed by safety fences, use a photoelectric sensor, a safety light curtain, etc. to make sure that the Robot stops its operation if the operator enters its operating range.

Failure to observe this Danger notice will result in death or serious injury due to contact with the Robot.

All personnel working with the Robot (safety administration, installation, operation, and maintenance personnel) must always be prepared and “Safety First” minded, to ensure the safety of all personnel.

WARNING

- In the vicinity of the area where the Robot is installed, avoid any dangerous actions, such as entering the Robot's operating range without due care.

Failure to observe this instruction may cause contact with the Robot or peripheral equipment, which may result in death or serious injury.

- Strictly observe the safety precautions and signs in the factory, such as “Flammable”, “High Voltage”, “Danger”, “Off-limits to Unauthorized Personnel”.

Failure to observe this instruction may result in death or serious injury do to fire, electric shock, caused by contact with the Robot or other equipment.

- Strictly observe the following precautions about clothing:
  - Always wear approved work clothes (no loose-fitting clothes).
  - To prevent mis-operation, do not wear gloves when operating the Robot.
  - Do not allow underwear, shirts, or neckties hang out from the work clothes.
  - Do not wear accessories, such as earrings, rings, or necklaces.
  - Always wear protective safety equipment, such as hard hats, safety shoes (with slip-proof soles), face shields, safety glasses, and gloves as necessary.

Failure to observe this instruction may result in death or serious injury.

- The following must be understood and strictly observed by all personnel as rules:
  - Unauthorized personnel other than the operator must not approach the area where the Robot is installed.

Failure to observe this instruction may cause contact with the Robot, Controller, control panel, workpiece, or Positioner, etc., may result in death or serious injury.

WARNING

- Perform the following inspection procedures prior to conducting manipulator teaching. If there is any problem, immediately take necessary steps to solve it, such as maintenance and repair.
  - Check for a problem in manipulator movement.
  - Check for damage to insulation and sheathing of external wires.
- Always return the programming pendant to the hook on the controller cabinet after use.

If the programming pendant is left unattended on the manipulator, on a fixture, or on the floor, etc., the Enable Switch may be activated due to surface irregularities of where it is left, and the servo power may be turned ON. In addition, in case the operation of the manipulator starts, the manipulator or the tool may hit the programming pendant left unattended, which may result in personal injury and/or equipment damage.

CAUTION

- All operators, programmers, maintenance personnel, supervisors, and anyone working near the system must be familiar with the operation of this equipment.
  - All personnel involved with the operation of the equipment must understand potential dangers of operation.
- General safeguarding tips:
  - Place system in E-STOP mode whenever it is not in use.
  - Use lockout/tagout procedures during equipment maintenance in accordance with ANSI/RIA R15.06-2012, section 4.2.5, Sources of Energy. Refer also to Section 1910.147 (29CFR, Part 1910), Occupational Safety and Health Standards for General Industry (OSHA).
  - Only trained personnel familiar with the operation of this equipment, the operator's manuals, the system equipment, and options and accessories can operate equipment.

Improper operation can result in personal injury and/or damage to the equipment.

- If the light in the operator's working space is not bright enough, provide the space with appropriate lighting.

CAUTION

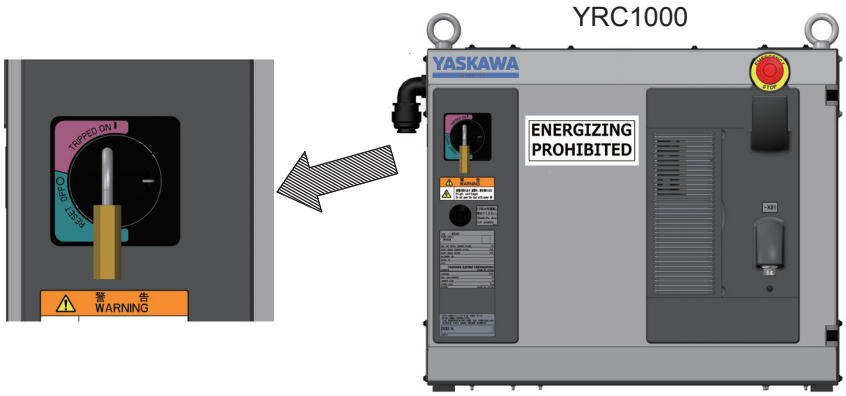
- Store industrial tools, etc. in a safe location outside the Robot's operating range.

If an industrial tool, etc. is left unattended on the Robot, on a fixture, or on the floor, etc., the Robot may come in contact with the industrial tool left unattended, which may result in damage to the Robot and/or the fixture.

Operation Safety

DANGER

- Personnel engaged in teaching or inspection, etc. of the Robot must receive special training required by applicable laws and regulations.
- While performing inspection and maintenance, wiring, or attaching a tool to the Robot, etc., make sure to turn OFF the power supply of the Controller and the tool, and keep the switch of the power supply locked so that unauthorized personnel cannot turn ON the power supply.
  In addition, display a warning sign stating “Energizing Prohibited”.



Turning ON the power supply without due care during inspection and maintenance, etc., may cause electric shock or unexpected movement of the Robot, which may result in personal injury.

- Use the Robot only within the specifications described in the manuals supplied with the Robot.

Failure to observe this instruction may result in personal injury and/or equipment damage.

- Observe the following precautions when performing a teaching operation within the Robot's operating range:
  - Be sure to perform lockout by putting a lockout device on the safety fence when going into the area enclosed by the safety fence. In addition, the operator of the teaching operation must display the sign that the operation is being performed so that no other person closes the safety fence.
  - View the Robot from the front whenever possible.
  - Always follow the predetermined operating procedure.
  - Always keep in mind emergency response measures against the Robot’s unexpected movement toward a person.
  - Ensure a safe place to retreat in case of emergency.

Failure to observe this instruction may cause improper or unintended movement of the Robot, which may result in personal injury.

DANGER

- Before operating the Robot, make sure servo power is OFF by performing the following operations. When servo power is OFF, the SERVO ON LED on the Pendant is OFF.
  - Press the EMERGENCY STOP buttons.
  - Disconnect the safety plug of the safety fence. (when in the PLAY mode or REMOTE mode)

If operation of the Robot cannot be stopped in an emergency, personal injury and/or equipment damage may result.

- Make sure that all safety protection devices are activated before starting a job in the PLAY mode.
- Confirm no person is present in the Robot's operating range and that the operator is in a safe location before:
  - Turning ON the Controller
  - Moving the Robot using the Pendant
  - Running the system in Manual mode
  - Performing automatic operations

Personal injury may result if a person enters the Robot's operating range during operation

- Immediately press an EMERGENCY STOP button whenever there is a problem.

WARNING

- Read “Safety” of the Controller instructions before operating.

Not reading and understanding chapter 1 of the Controller instruction can result in death or serious injury.

- Read and understand all Warning Labels before operating.

Not reading and understanding all Warning Labels can result in death or serious injury.

- Confirm that no person is present in the P-point maximum envelope of the Robot before:
  - Turning on the power for the Controller.
  - Moving the Robot with the Programming Pendant.
  - Running the system in Manual mode.
  - Performing automatic operations.

Injury may result if anyone enters the working envelope of the Robot during operation. Always press an EMERGENCY STOP button immediately if there are problems.

- Observe the following when performing teaching operation within the operating range:
  - Lockout by putting a lockout device on the safety fence when going into the area enclosed by the safety fence.
  - Display a sign that operations are being performed so no other person closes the safety fence.
  - View from the front whenever possible.
  - Always follow the predetermined operating procedure.
  - Always keep in mind emergency response measures against unexpected movement toward a person.
  - Ensure a safe place to retreat in case of emergency.

Failure to observe this precautions may cause improper or unintended movement, which may result in personal injury.

Maintenance Safety

WARNING

- Use care when modifying software.
  - The equipment allows modifications to the software for maximum performance.

All modifications made to the software will change the way the equipment operates and may cause death or serious injury, as well as damage parts of the system.

- Make sure all modifications did not make create a hazardous or dangerous condition in all modes.

All modifications made to the software will change the way the equipment operates and may cause death or serious injury, as well as damage parts of the system.

- Disconnect and lockout/tagout all sources of energy before making modifications or connections.

Not disconnecting and doing lockout/tagout of all sources of energy can result in death or serious injury.

CAUTION

- Do not modify the Controller.

Making modifications without written permission from YASKAWA voids the warranty.

- Back up all programs and jobs onto suitable media before program changes are made.

To avoid loss of information, programs, or jobs, a backup must always be made before any service procedures are done and before any changes are made to options, accessories, or equipment.

- All connections must be made within the standard voltage and current ratings of the equipment.

Improper connections can damage the equipment.

Definition of Terms Used Often in This Manual

The Robot is the YASKAWA industrial robot product.

The Robot usually consists of a Robot, Controller, Programming Pendant, and Robot cables.

In this manual, the equipment is designated as follows:

Equipment	Manual Designation
MLX300/YRC1000micro Controller MLX300/YRC1000 Controller or MLX300/DX200 Controller	Robot Controller
YRC1000micro programming pendant safety signal short circuit connector	Programming pendant safety signal short circuit connector (optional)
YRC1000micro, YRC1000 or DX200 Programming Pendant	Robot Programming Pendant
Proface or Allen Bradly MobileView	PLC Pendant
Conveyor Tracking Option Manipulator	Robot
Cable between the Robot and the Controller	Robot cable
Positioner	Positioner

Descriptions of the Robot Programming Pendant keys, buttons, and displays are shown as follows:

Equipment		Manual Designation
Programming Pendant	Character Keys /Symbol Keys	The keys which have characters or symbols printed on them are denoted with [ ]. e.g. [ENTER]
	Axis Keys /Numeric Keys	[Axis Key] and [Numeric Key] are generic names for the keys for axis operation and number input.
	Keys pressed simultaneously	When two keys are to be pressed simultaneously, the keys are shown with a “+” sign between them, e.g. [SHIFT]+[COORD].
	Mode Switch	Mode Switch can select three kinds of modes that are denoted as follows: REMOTE, PLAY or TEACH. (The switch names are denoted as symbols)
	Button	The three buttons on the upper side of the Programming Pendant are denoted as follows: START, HOLD, or EMERGENCY STOP. (The button names are denoted as symbols)
	Displays	The menu displayed in the Programming Pendant is denoted with { }. e.g. {JOB}

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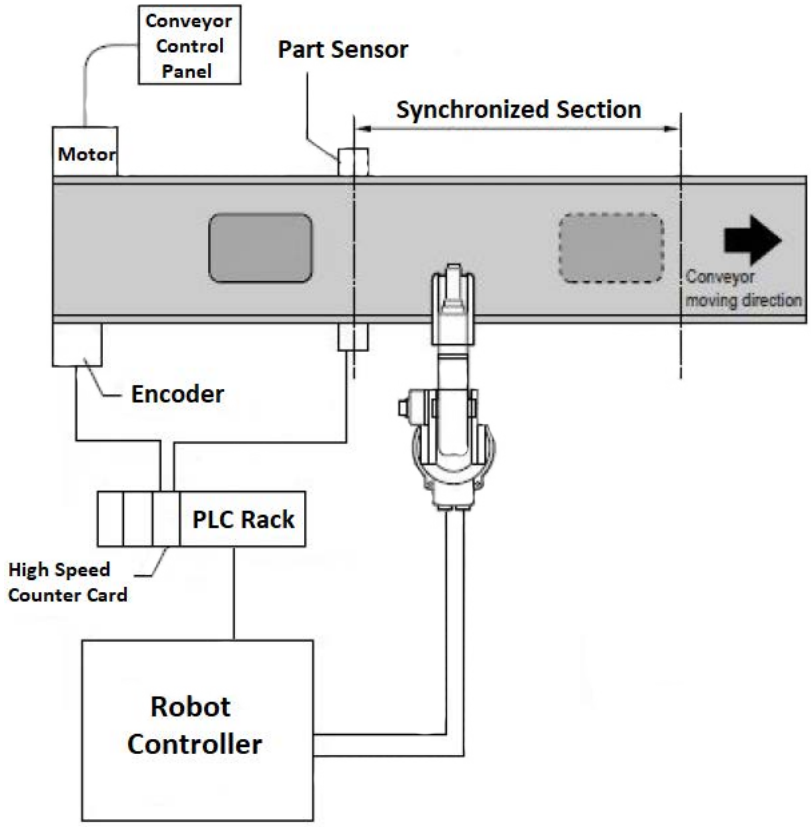
1 Introduction

This manual describes the configuration and use of the MLX300 Conveyor Tracking feature. The Add On Instruction (AOI)s and other functionality listed in this manual are only available if the Conveyor Tracking option is enabled on the Robot Controller.

1.1 Conveyor Tracking Overview

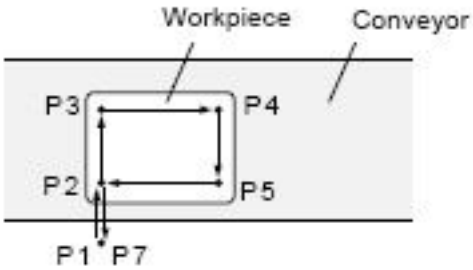
The conveyor tracking AOIs enable the robot to perform movements relative to a part as it travels down a conveyor. The positions are originally taught with the conveyor stopped. During playback, any motion commands between the MLxRobotConvSyncStart and MLxRobotConvSyncStop AOIs will be synchronized with the linear motion of the part moving on the conveyor.

Fig. 1-1: Conveyor Tracking Cell Configuration



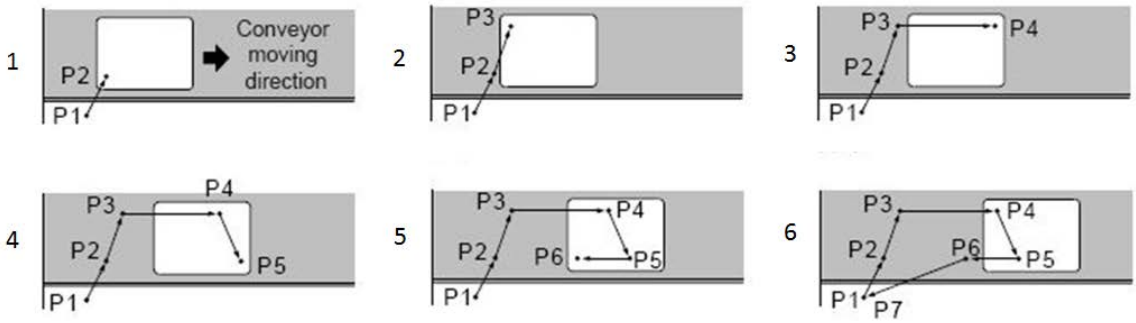
Referring to the example in *Fig. 1-2 "Robot Taught Positions with Conveyor Off"*, the conveyor stops with the part at a known distance from the sensor. Seven points are taught near the part. The move commands points P2-P6 be placed in the job between the MLxRobotConvSyncStart and MLxRobotConvSyncStop AOIs.

Fig. 1-2: Robot Taught Positions with Conveyor Off



In Automatic mode, when the job executes, the robot moves to the points taught relative to the part, while moving down the conveyor. The conveyor motion will be interpolated or added to the taught positions to create a tracking sequence (*Fig. 1-3 "Robot Path During Playback"*.)

Fig. 1-3: Robot Path During Playback



1.2 Conveyor Tracking Requirements

The following items are needed for conveyor tracking:

- **MLX300 Controller**
- **Conveyor**
- **Incremental Encoder** - mounted to the conveyor for reading conveyor position.
- **Counter Card** - reads encoder position with ControlLogix/ CompactLogix.
- **Sensor or Camera with a Trigger Output** - captures the position of an object on the conveyor. Normally, this output is wired in to the “latch” input of the counter card.

NOTICE

- To improve conveyor tracking accuracy, use a thru-beam/opposed mode sensor.
- Instead of using the encoder and counter card, there is an option to use the drive feedback over the Ethernet.

1.2.1 Equipment Used for Verifying Conveyor Tracking Operations

Conveyor tracking has been tested using the following:

- **Conveyor** - A straight conveyor with speeds up to 1000 mm/sec.
- **Incremental Encoder** - Allen Bradley 845H
- **High Speed Counter Card (HSC)** - The 1756-HSC/B Version 3.4 card for ControlLogix and the 1769-HSC/B Version 2.1 card for CompactLogix.
- **Sensor** - Photo Eye/Switch (e.g. Keyence # FS-N41P or Banner # PIT46U sensor, along with the correct fibers for the conveyor width)

1.2.2 RSLogix 5000 files included with the MLX300 for Conveyor Tracking

The controller is delivered with example conveyor tracking programs which can be used as a starting point to build a full conveyor tracking application.

- **Single Robot Tracking Programs**
  - **MLxConveyorSync_0.L5X** - Program to sync the conveyor to a single robot
  - **ConveyorTrackingExample.L5X** - Example Conveyor Tracking program for a single robot
- **Dual Robots Tracking Programs**
  - **MLxConveyorSync_0_R0R1.L5X** - Program to sync the conveyor to two robots on the same controller
  - **ConveyorTrackingExample_R0R1.L5X** - Example Conveyor Tracking program for R0 and R1
  - **MLxConveyorSync_0_C0C1.L5X** - Program to sync the conveyor to two robots on different controllers
  - **ConveyorTrackingExample_C0C1.L5X** - Example Conveyor Tracking program robots on C0 and C1
- **Options for Simulation or Drive Position Feedback**
  - **Main_for_Sim_or_Drive_FB.L5X** - An optional Main routine for the Sync program to use a simulated position generator or the Drives Feedback instead of an encoder and counter card.
- **Conveyor Tracking AOIs**
  - **MLxRobotConvSyncStart.L5X** - Start conveyor sync
  - **MLxRobotConvSyncStop.L5X** - Stop conveyor sync
  - **MLxRobotConvSyncStopWithAxisMot.L5X** - Stop conveyor sync and bend into axis move
  - **MLxRobotConvSyncStopWithLinMot.L5X** - Stop conveyor sync and bend into linear move

1.3 Manual References

There may be a requirement for the following manuals when configuring the Conveyor Tracking Option.

Table 1-1: Reference Manuals

Part Number	Description
180952-1CD	*MLX300 Hardware Installation, Software Upgrade and Options*
180247-1CD	*MLX300 Software and Operating User's*
188423-1CD	*Ethernet/IP-Safe Configuration, MLX300 Controller as Adapter Supplement*

1.4 YASKAWA Representative Information

The following information is for a YASKAWA representative in America, for other locations refer to the back cover.

(937) 847-3200

For **routine** technical inquiries, you can also contact a YASKAWA representative at the following e-mail address:

techsupport@motoman.com

When using e-mail to contact a YASKAWA representative, please provide a detailed description of your issue, along with complete contact information. Please allow approximately 24 to 36 hours for a response to your inquiry.

NOTICE

Please use e-mail for routine inquiries only. If you have an urgent or emergency need for service, replacement parts, or information, you must contact a YASKAWA representative at the telephone number shown above or on the back cover.

Please have the following information ready before calling a YASKAWA representative:

• System	Conveyor Tracking Option
• Manipulator	
• Positioner	
• Primary Application	
• Robot Controller	
	MLX300/YRC1000micro, MLX300/YRC1000 or MLX300/DX200
• Software Version	
	Access this information on the Programming Pendant's LCD display screen by selecting {MAIN MENU} - {SYSTEM INFO} - {VERSION}
• Robot Serial Number	
	Located on the Manipulator data plate
• Robot Sales Order Number	
	Located on the Robot Controller data plate
• Warranty ID:	
	Located on Robot Controller

2 Configuring Conveyor Tracking

This section explains how to setup the conveyor tracking tags and the High Speed Counter (HSC) card for the CompactLogix PLC.

NOTICE

Use this section as a reference for setting up the ControlLogix PLC HSC card.

2.1 Configuring the HSC Card

This section describes how to set up the HSC card (1769-HSC) for the CompactLogix PLC. The setup for the ControlLogix HSC card is similar, and this section is available as a reference.

NOTICE

The ControlLogix HSC card is not exactly the same as the CompactLogix PLC, but the information on the screens is the same.

2.1.1 Wiring the HSC Card

There are several ways to configure the HSC card. Each HSC card can support up to four independent counters. This document uses only one of the four counters.

Connect each conveyor encoder to one counter input. Wire the HSC is the following manner:

- HSC card Input A: Wired to conveyor encoder channel A
- HSC card Input B: Wired to conveyor encoder channel B
- HSC card Input Z: Wired to sensor switch or any other input that acts as a trigger for object detection, such as a vision system.

Refer to the HSC USERS GUIDE for details on wiring the HSC card to a conveyor.

NOTICE

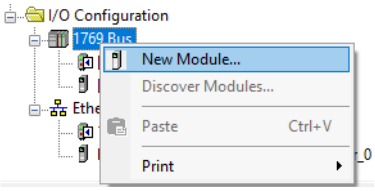
Use Through-Beam Sensors, where a part breaks a through-beam, for better tracking accuracy.

2.1.2 Configuring the HSC Card in Studio 5000

2.1.2.1 Adding a HSC Card to the CompactLogix PLC

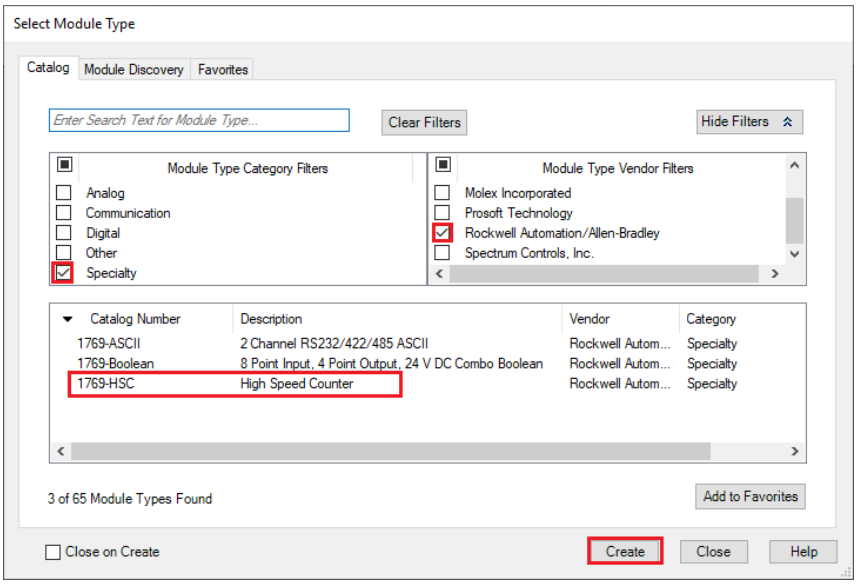
1. Right-click on the “1769 Bus” in the “I/O Configuration”, select “New Module.” and navigate to the HSC card.

Fig. 2-1: Adding a New Module



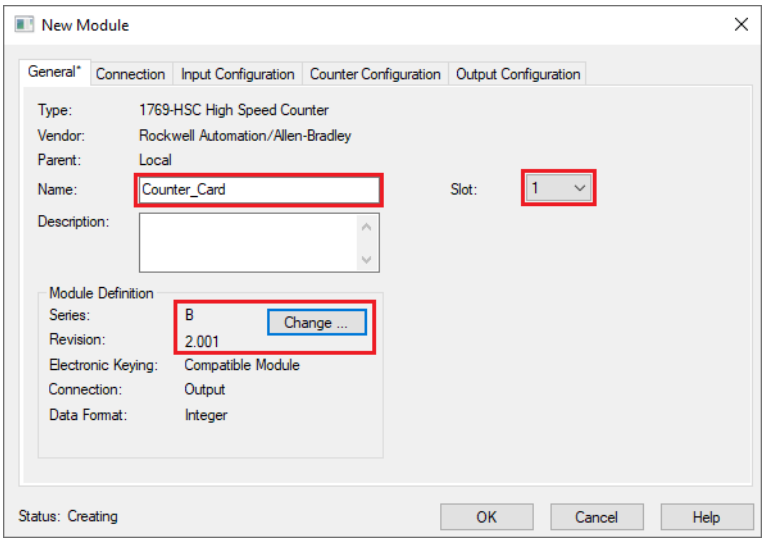
2. Using the check boxes, select “Specialty”, Rockwell Automation/ Allen-Bradley and 1769-HSC and press {Create}.
See *Fig. 2-2 “Adding the 1769-HSC Card to PLC Rack”*.

Fig. 2-2: Adding the 1769-HSC Card to PLC Rack



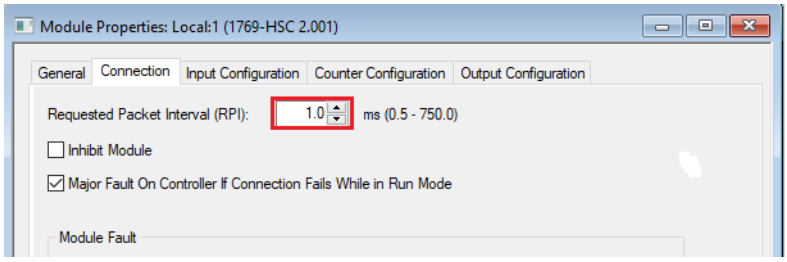
3. On the {General} tab, enter the “Name:” and set the “Slot:” number for the HSC.

Fig. 2-3: 1769-HSC General Tab



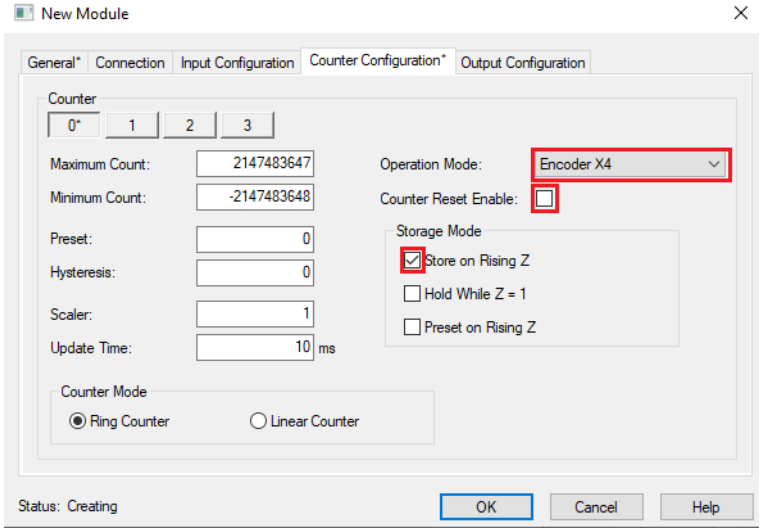
4. Make sure the “Series” and “Revision” numbers are correct for the actual card being used.
5. Click on the {Connection} tab, and set the “Requested Packet Interval (RPI)” to 1.0 ms.

Fig. 2-4: 1769-HSC Connection Tab



6. Make sure that “Inhibit Module” is unchecked and “Major Fault on Controller if Connection Fails While in Run Mode” is checked.
7. Click on the {Counter Configuration} tab, and select “Encoder X4” for the “Operation Mode”.

Fig. 2-5: CompactLogix 1769-HSC Counter Configuration Tab

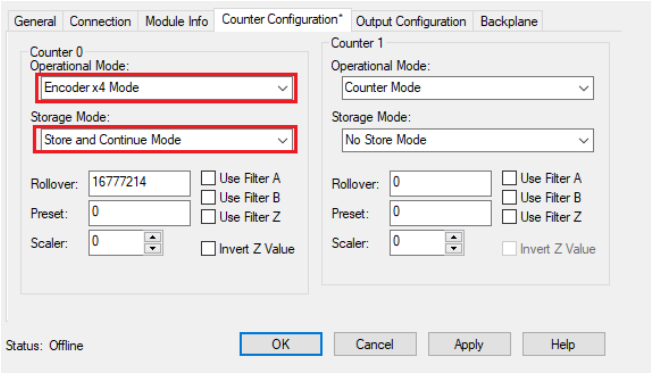


8. Uncheck the “Counter Reset Enable” box, check the “Store on Rising Z” box, and press {OK}.
9. Press {Close} on the “Select Module Type” screen.

2.1.2.2 Adding a HSC Card to the ControlLogix PLC

The steps in *section 2.1.2.1* for an CompactLogix also apply to the ControlLogix PLC, with the exception that the Counter Configuration screen is slightly different. refer to *Fig. 2-6*

Fig. 2-6: ControlLogix 1756-HSC Counter Configuration Tab



1. Set Operation Mode to “Encoder x4 Mode”
2. Set Storage Mode to “Store and Continue Mode”

2.2 Importing Conveyor Tracking Programs

WARNING

- Failure to set up the Conveyor Tags correctly can lead to unexpected behavior during conveyor tracking.

Not setting the Conveyor Tags correctly may result in death or serious injury

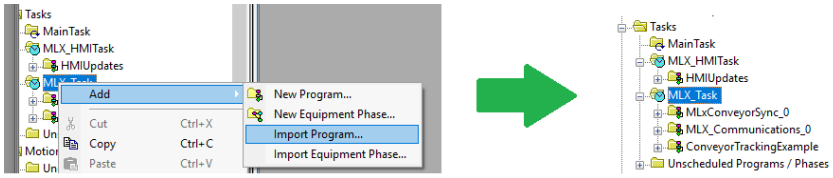
NOTICE

- Importing the example program also imports the conveyor tracking AOIs.
- Ignore any import errors due to the HSC card being in a different slot.
  - For a new ladder configuration, do not overwrite any tags, data types, or AOIs.

1. Follow the steps below to import the conveyor sync and example programs from one of the following folders on the MLX Software CD under the PLC ➡ Conveyor Tracking folder.
   - Single Robot Tracking Programs
   - Dual Robots Tracking Programs
     - Two Robots Tracking On Same Controller
     - Two Robots Tracking On Different Controllers

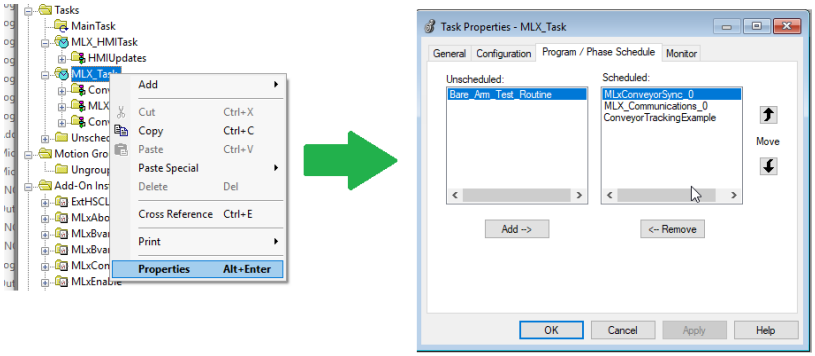
2. Right-click on the “MLX_Task, press “Add,” and choose “Import Program.”

Fig. 2-7: Importing Conveyor Tracking Programs



3. Select each of the following programs for single robot tracking:
   - MLxConveyorSync_0.L5X
   - ConveyorTrackingExample.L5X
4. Right-click MLX_Task, and select “Properties”.

Fig. 2-8: Scheduling Program Order



5. On the “Program / Phase Schedule” tab, use the [Move] buttons to order the programs as shown on the right side of *Fig. 2-8*.
6. After importing the conveyor tracking programs, several new tags will show up in the PLC Controller tags.

Fig. 2-9: Controller Tags for Conveyor Tracking

⊞ Conveyor0_CurrentValue	DINT
Conveyor0_CurrentValue_mm	REAL
Conveyor0_EncoderEnable	BOOL
Conveyor0_EncoderTriggerDisable	BOOL
⊞ Conveyor0_LatchedValue	DINT
Conveyor0_LatchedValue_mm	REAL
Conveyor0_NewData	BOOL
Conveyor0_ResetCounter	BOOL
Conveyor0_ResetNewData	BOOL
Conveyor0_Rollover	REAL
Conveyor0_SensorState	BOOL

NOTICE

- These tags will have the prefix “Conveyor0” for the first conveyor, “Conveyor1” for the second conveyor, etc…
- These tags need to be linked to the proper hardware devices being used for reading/latching the conveyor position.
- *Table 2-1 “Conveyor Tracking Tag Descriptions”* shows a brief introduction to what each of these tags means.

Table 2-1: Conveyor Tracking Tag Descriptions

Tag Name	Input/ Output	Description
Conveyor0_SensorState	Input	This bit should turn on when an object is in front of the sensor or under a camera.
Conveyor0_NewData	Input	This bit should turn on when a new object has passed by the camera and stay on until the ResetNewData signal is sent
Conveyor0_ResetCounter	Output	Turning this bit on should reset the counter card position to 0.
Conveyor0_ResetNewData	Output	Turning this bit on will reset the NewData flag (used internally to tell the counter that the current object has been processed and queued).
Conveyor0_CurrentValue	Input	This should contain the current position of the conveyor in encoder counts.
Conveyor0_LatchedValue	Input	This should contain the current latched position of the conveyor. When a new object passes the sensor, this position will be stored until the ResetNewData flag is sent.
Conveyor0_RollOver	Input	The RollOver value of the conveyor. This must be set to the proper RollOver value or Conveyor Tracking will not function properly during rollover conditions.
Conveyor0_EncoderEnable	Output	This tag enables the encoder processing. This tag is not used for the ControlLogix HSC card.
Conveyor0_EncoderTrigger Disable	Output	This tag will disable the sensor input. This tag is not used for the ControlLogix HSC card,

NOTICE

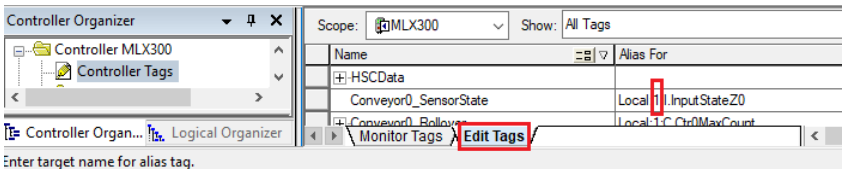
This manual uses “Conveyor0” prefix to describe the setup, configuration, and operation for Conveyor Tracking. If using a different conveyor number, simply replace with the correct conveyor number in the prefix.

2.3 Linking the Conveyor Tags to the HSC Card Tags

After configuring the HSC card, the Conveyor Tags need to be linked to the proper HSC tags as shown in *Fig. 2-11(a "CompactLogix: Conveyor Tags Linked to HSC"* and *Fig. 2-11(b "ControlLogix: Conveyor Tags Linked to HSC"*.

1. Press the "Edit Tags" tab on the "Controller Tags" screen.

Fig. 2-10: Controller Tag Screen



2. Enter the data shown in *Fig. 2-11(a)* and *Fig. 2-11(b)* in the middle column "Alias For", for each new tag in the left column

NOTICE

The "Local" number needs to be set to the slot the HSC card is installed in. For this example, it is slot 1

Fig. 2-11(a): CompactLogix: Conveyor Tags Linked to HSC

Conveyor0_CurrentValue	Local:1:I.Ctr0CurrentCount	L	DINT
Conveyor0_CurrentValue_mm			REAL
Conveyor0_EncoderEnable	Local:1:O.Ctr0En	L	BOOL
Conveyor0_EncoderTriggerDisable	Local:1:O.Ctr0ZInhibit	L	BOOL
Conveyor0_LatchedValue	Local:1:I.Ctr0StoredCount	L	DINT
Conveyor0_LatchedValue_mm			REAL
Conveyor0_NewData	Local:1:I.Ctr0RisingEdgeZ	L	BOOL
Conveyor0_ResetCounter	Local:1:O.Ctr0SoftPreset	L	BOOL
Conveyor0_ResetNewData	Local:1:O.Ctr0ResetRisingEdgeZ	L	BOOL
Conveyor0_Rollover	Local:1:C.Ctr0MaxCount	L	DINT
Conveyor0_SensorState	Local:1:I.InputStateZ0	L	BOOL

Fig. 2-11(b): ControlLogix: Conveyor Tags Linked to HSC

Conveyor0_CurrentValue	Local:2:I.PresentValue[0]	L	DINT
Conveyor0_CurrentValue_mm			REAL
Conveyor0_EncoderEnable			BOOL
Conveyor0_EncoderTriggerDisable			BOOL
Conveyor0_LatchedValue	Local:2:I.StoredValue[0]	L	DINT
Conveyor0_LatchedValue_mm			REAL
Conveyor0_NewData	Local:2:I.NewDataFlag.0	L	BOOL
Conveyor0_ResetCounter	Local:2:O.ResetCounter.0	L	BOOL
Conveyor0_ResetNewData	Local:2:O.ResetNewDataFlag.0	L	BOOL
Conveyor0_Rollover	Local:2:C.RollOver[0]	L	DINT
Conveyor0_SensorState	Local:2:I.ZState.0	L	BOOL

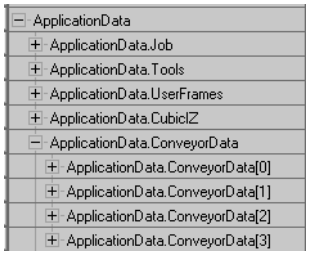
NOTICE

The Conveyor0_CurrentValue, Conveyor0_LatchedValue, and Conveyor0_RollOver are in units of pulse counts.

2.4 Conveyor Tag Configuration for the Robot Controller

The information for configuring a specific conveyor for the Robot Controller is located in the Controller Tags under the ApplicationData.ConveyorData tag structure.

Fig. 2-12: Conveyor Data Inside Application Data



The ConveyorData tag structure contains the following variables:

Fig. 2-13: Conveyor Application Data

ApplicationData.ConveyorData	{...}
ApplicationData.ConveyorData[0]	{...}
ApplicationData.ConveyorData[0].ConveyorName	'Infeed Conveyor'
ApplicationData.ConveyorData[0].IsActive	1
ApplicationData.ConveyorData[0].ControllerIndex	0
ApplicationData.ConveyorData[0].ConveyorPosition	376.0474
ApplicationData.ConveyorData[0].ConveyorSpeed	250.0
ApplicationData.ConveyorData[0].ConveyorStartPosition	{...}
ApplicationData.ConveyorData[0].ConveyorTeachPosition	{...}
ApplicationData.ConveyorData[0].MaxStartPosition	{...}
ApplicationData.ConveyorData[0].UserFrameNumber	10
ApplicationData.ConveyorData[0].ConveyorType	0
ApplicationData.ConveyorData[0].EncoderToMMConversion	0.0281958
ApplicationData.ConveyorData[0].SpeedConversionFactor	0.0
ApplicationData.ConveyorData[0].NbrOfPointsToLinearFit	25
ApplicationData.ConveyorData[0].NbrOfPointsToAverage	25
ApplicationData.ConveyorData[0].LagOffset	0.0

Table 2-2: Conveyor Tag Descriptions

Parameter Name	Description
Conveyor Name	User defined name of conveyor.
IsActive	Turns conveyor reading on for conveyor. This must be turned on for any active conveyors and turned off for unused conveyors to preserve MLX_Task load.
ControlIndex	Defines the Robot Controller being used if PLC is connected to multiple Robot Controllers.
ConveyorPosition	Feedback conveyor position in mm (Note this value is for reading only - modifying this value has no effect.)
ConveyorStartPosition	Conveyor Position where tracking operation begins (distance past sensor). This is an array of four positions. There is a start position for up to four robots.
ConveyorTeachPosition	Conveyor Position where positions are taught (distance past sensor) This is an array of four positions. There is a teach position for up to four robots.
MaxStartPosition	Conveyor Position where the part should be ignored, because it moves out of reach, or OL - overlimit (distance past sensor) This is an array of four positions. There is a max start position for up to four robots.
UserFrameNumber	User frame number taught for the conveyor. The conveyor movement is in the direction of +X. **NOTE**: The PLC job needs to set the active frame to this number using the MLxRototSetUserFrame instruction)
ConveyorType	Not Used. Set to 0.
EncoderToMMConversion	The number of mm per encoder count. i.e. [EncoderCount] * EncoderToMMConversion = Distance (mm))
SpeedConversionFactor	Not Used
NbrOfPointsToLinearFit	Not Used
NbrOfPointsToAverage	The number of points used to calculate a moving average of the encoder positions. Useful for noisy encoder signal. The valid range for this parameter is 0 - 100. Default setting is 25.
LagOffset	An offset in mm along the x direction of the conveyor. This can offset a constant error along the conveyor. This value can be used to fine tune the execution positions of the motion instructions that execute after a call to MLxRobotConvSyncStart. A positive value corresponds to adjustments in the +X direction.

2.5 Task Period and Jog Timer Adjustments

The complete configuration of the PLC network needs to be considered. The addition of robots, and other network devices could cause Ethernet communication issues with the Robot Controllers. Depending on the actual load of the PLC, the settings in this section might need to be changed.

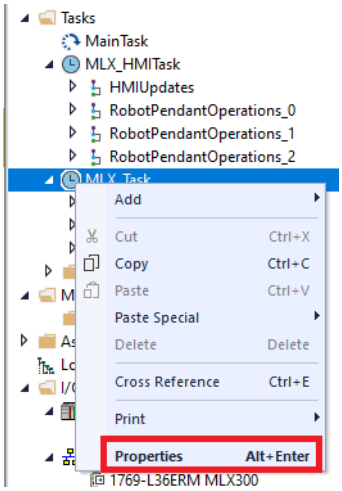
2.5.1 Adjusting Task Periods

The default MLX_HMITask Period is set to 16 ms. For large PLC programs the period may need to be increased to allow more processing time for other devices. YASKAWA recommends keeping the 16 ms setting, unless there are issues.

Change the MLX_Task Period to 8 ms

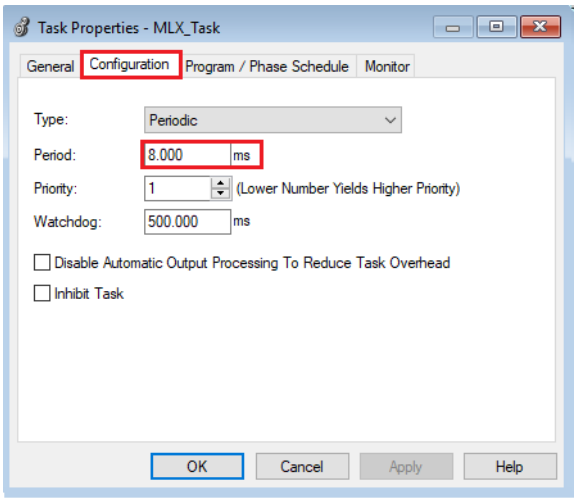
1. Right-click on “MLX_Task”, and select “Properties”

Fig. 2-14: Changing MLX_Task Properties



2. Select the “Configuration” tab.

Fig. 2-15: Increasing Period of MLX_Task



3. Increase the “Period” from 4.000 ms to 8.000 ms and press {OK}.

2.6 Conveyor Tracking Setup Procedure

2.6.1 Verify Counter Card is Functional

WARNING

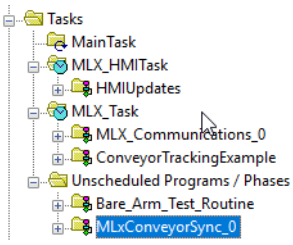
- Make sure Conveyor Tracking Tags match operations.

Not making sure Conveyor Tracking Tags match operations will not work correctly and may cause unexpected robot behavior which may result in death or serious injury.

Before attempting Conveyor Tracking operations, the functionality of the counter card (or comparable device) should be verified.

1. Click on and slide the MLxConveyorSync_0 program into the “Unscheduled Programs” folder to unschedule.

Fig. 2-16: Unschedule MLxConveyorSync_0



2. Access the “Conveyor0_” tags in the Controller Tags list.
3. Set “Conveyor0_EncoderEnable” to 1 for the CompactLogix HSC card.

Fig. 2-17: Conveyor0 Tags

Conveyor0_SensorState	0
+ Conveyor0_Rollover	2147483647
Conveyor0_ResetNewData	0
Conveyor0_ResetCounter	0
Conveyor0_NewData	1
+ Conveyor0_LatchedValue	75954
Conveyor0_EncoderTriggerDisable	0
Conveyor0_EncoderEnable	1
+ Conveyor0_CurrentValue	209965

NOTICE

The ControlLogix HSC card does not have the “Conveyor0_EncoderEnable” tag.

4. Turn conveyor on and start moving. The Conveyor0_CurrentValue should start increasing.

NOTICE

If the Conveyor0_CurrentValue value decreases;

- Reverse the encoder direction by changing Local:x:O.Ctr0DirectionInvert, where “x” is the slot number.

OR

- Swap the wires to the A and B channels of the HSC card.

5. Place an object on the conveyor and move it past the sensor.
   - As it crosses, the Conveyor0_CurrentValue stores in the Conveyor0_LatchedValue tag, and the Conveyor0_NewData bit turns to 1.
   - When the part is blocking the sensor The Conveyor0_SensorState will be 1.
6. Toggle the Conveyor0_ResetNewData bit.
   - The Conveyor0_NewData bit turns to 0.

NOTICE

- If the counter card does not work properly it is important to debug the counter card before moving on.
- The Conveyor0_ResetCounter can be toggled to reset the Conveyor0_CurrentValue to 0.

7. Re-schedule the MLx_ConveyorSync_0 program under the MLX_Task and move it to the top of all scheduled programs.

2.6.2 Calculate Conveyor Resolution

1. Access the ApplicationData.ConveyorData[0] tags. These tags were explained in *section 2.4 “Conveyor Tag Configuration for the Robot Controller”*.

Fig. 2-18: Conveyor Application Data

ApplicationData.ConveyorData	{...}
ApplicationData.ConveyorData[0]	{...}
ApplicationData.ConveyorData[0].ConveyorName	'Infeed Conveyor'
ApplicationData.ConveyorData[0].IsActive	1
ApplicationData.ConveyorData[0].ControllerIndex	0
ApplicationData.ConveyorData[0].ConveyorPosition	376.0474
ApplicationData.ConveyorData[0].ConveyorSpeed	250.0
ApplicationData.ConveyorData[0].ConveyorStartPosition	{...}
ApplicationData.ConveyorData[0].ConveyorTeachPosition	{...}
ApplicationData.ConveyorData[0].MaxStartPosition	{...}
ApplicationData.ConveyorData[0].UserFrameNumber	10
ApplicationData.ConveyorData[0].ConveyorType	0
ApplicationData.ConveyorData[0].EncoderToMMConversion	0.0281958
ApplicationData.ConveyorData[0].SpeedConversionFactor	0.0
ApplicationData.ConveyorData[0].NbrOfPointsToLinearFit	25
ApplicationData.ConveyorData[0].NbrOfPointsToAverage	25
ApplicationData.ConveyorData[0].LagOffset	0.0

2. Set the ApplicationData.ConveyorData[0].IsActive to 1.
3. With the conveyor stopped, place a part on the conveyor.
4. Place a ruler next to the conveyor and record the Conveyor0_CurrentValue for the current position. (Initial Pulse Count)
5. Operate the conveyor to move the part approximately 1000 mm.
6. Measure the distance that the part traveled (mm) and record the new Conveyor0_CurrentValue (Final Pulse Count).
7. Calculate the pulse count conversion factor
   – Conversion Factor = Distance (mm) / (Final Pulse Count - Initial Pulse Count).
8. Enter the Conversion Factor in ApplicationData.ConveyorData[0].EncoderToMMConversion tag.
9. Verify the Conversion Factor is correct,
10. Operate the conveyor again and verify that the changing ApplicationData.ConveyorData [].ConveyorPosition variable is reporting the correct distance traveled in mm.
    – When parts move past the sensor, the conveyor position sets to 0.
    – After part moves 1000 mm past the sensor, stop the conveyor and verify the ApplicationData.ConveyorData [].ConveyorPosition value is the correct distance.

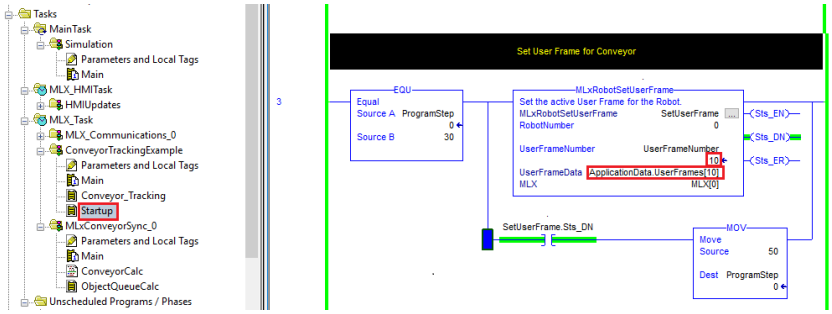
NOTICE

- The switch must be in Manual mode for ApplicationData.ConveyorData [].ConveyorPosition to reset to 0 when the part crosses the sensor.
- If the switch is in Automatic, the ApplicationData.ConveyorData [].ConveyorPosition will reset to 0 when the queue is empty.

2.6.3 Teach a User Frame

1. Install a pointing device on the robot and enter the tool data.
2. Teach a user frame for the conveyor, so motion is in the +X direction of the user frame
3. Enter the frame number in the ApplicationData.ConveyorData[0].UserFrameNumber tag.
4. Change the user frame number in the MLxRobotSelectUserFrame command for the “Startup” routine.
   –An example program is shown in *Fig. 2-19*. This user frame number was changed to 10.

Fig. 2-19: Setting User Frame in Startup Routine



NOTICE

The MLxRobotSetUserFrame command is already in the ConveyorTrackingExample program. This AOIs user frame needs to be changed to the frame taught.

2.6.4 Teach Points and Setup Tracking Parameters

NOTICE

The Teach Points and Setup Tracking Parameters section needs to be completed for each robot number.

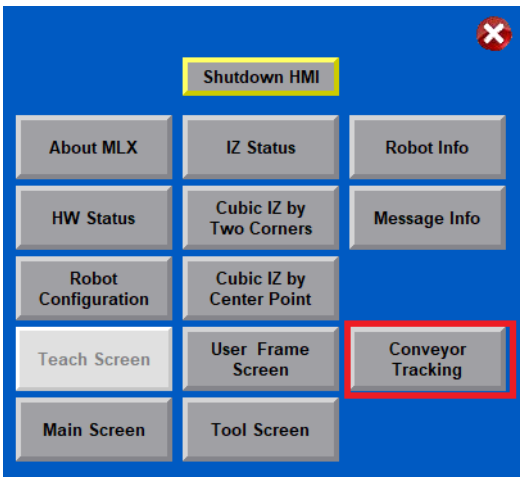
■ **Conveyor Teach Position Value**

WARNING

The motions performed while conveyor tracking will be different from those taught with the conveyor stationary. Thus, certain errors may be encountered while tracking that would not show up otherwise (e.g. speed/limit violations).

1. Press {Menu} then {Conveyor Tracking} on the Pendant.

Fig. 2-20: Selecting Conveyor Tracking



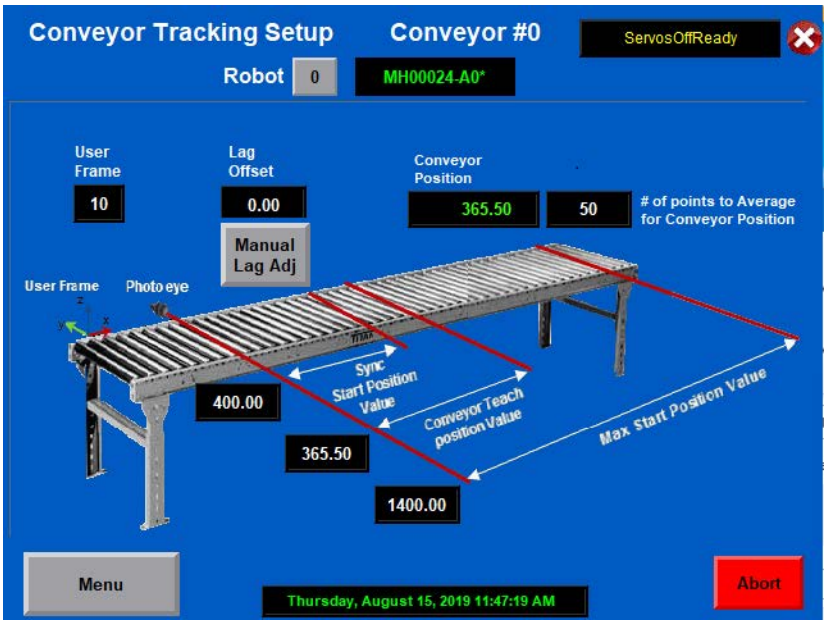
NOTICE

If the {Conveyor Tracking} button does not display on the Menu screen, it is due to not enabling the conveyor tracking license on the controller.

2. Select the robot number at the top of the screen
3. Press the button below the Lag Offset until it displays {Manual Lag Adj}.

NOTICE
When pressing this button the text on the button changes from {Auto Lag} to {Manual Lag Adj}, and back. So this button may need to be pressed to display {Manual Lag Adj}

Fig. 2-21: Conveyor Tracking Setup Screen



4. Turn the key switch to Manual mode.
   – Verify every time a part crosses the sensor the Conveyor Position reset to 0.
5. Operate the conveyor to move a part past the photo eye and stop the conveyor at a position where the part will be taught.
   – Refer to the Conveyor Position, shown on the HMI, and enter this number as the “Conveyor Teach Position Value”. This is the conveyor position where all the robot motion points will be taught.
6. Teach all robot positions relative to the part with the conveyor stopped.
   – The positions need to be taught in the World Coordinate system.
   – Typical points taught will be an approach position above the part, a grip position at the part, and a depart position above the part.
   – During playback, Linear Motion instructions can be used to move the robot TCP to these points while tracking the part motion on the conveyor, as long as they fall between a MLxRobotConvSyncStart and a MLxRobotConvSyncStop.
   – Only Linear move commands are used for conveyor tracking.
   – For a faster speed use the VMAX option of the motion commands.
   – All of the entries and feedback on the Pendant Conveyor Tracking screen are stored in the ApplicationData.ConveyorData[0] tags.

7. Enter the remaining parameters on the “HMI Conveyor Tracking Setup” screen:

■ **Sync Start Position Value**
Determine how far past the photo eye the conveyor tracking sequence should begin. The robot will wait at the MLxRobotConvSyncStart instruction until the part has traveled to this distance past the sensor. At this point, the robot will begin to move.

NOTICE
There are four values for the ConveyorStartPosition, ConveyorTeachPosition and MaxStartPosition in the ApplicationData.ConveyorData[0] tag array. These values can be used for up to four robots, so that they can work on different sections of the conveyor.

■ **Max Start Position Value**
Estimate the worst case distance past the photo eye that the robot will be able to begin tracking a part and successfully pick it up without reaching a software limit. This parameter is used for queued parts. After moving a part, if a new part had already crossed the photo eye and is in a queue, the MLxRobotConvSyncStart AOI will determine if its location is past the Max Start Position value. If the part is past the Max Start Position, the AOI will turn on it's Sts_OL output, (Over Limit). Whenever the speed of the conveyor is changed, this value should be adjusted.

■ **# of Points to Average for Conveyor Position**
The number of points used to calculate a moving average of the encoder positions. Useful for a noisy encoder signal.

■ **Conveyor Position**
This field is for conveyor position reference only and cannot be changed.

■ **Lag Offset**
The lag offset parameter can be adjusted to speed up or slow down the SyncStart instruction execution time, so that parts can be picked up at a consistent location on the part. It can compensate for I/O hardware delays. A positive value corresponds to an adjustment in the +X direction.

■ **Manual/Auto Lag Adj**
Use this button to manually enter a Lag Offset value, or to enable a lookup table to automatically change the Lag Offset based on the current speed. Refer to *section 2.7*.

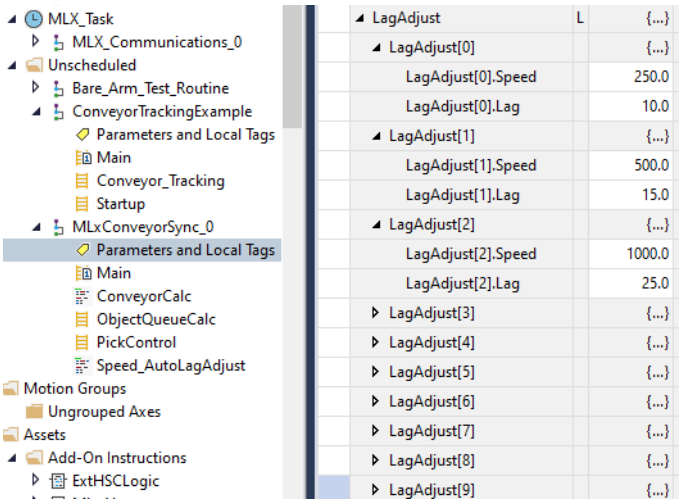
2.7 Setting Up the Automatic Lag Offset Adjustment

In general, a Lag Offset setting is only good for a range of conveyor speeds. When the conveyor speed changes, the robot might start picking the part ahead or behind the center of the part. To resolve this issue, use the Auto Lag Adj feature to change the Lag Offset, based on the conveyor speed feedback.

When turning on the “Auto Lag Adj”, a routine will automatically adjust the Lag Offset for different conveyor speeds. It uses the speed feedback value with a Speed/Lag Offset lookup table to decide the Lag Offset. The routine will interpolate the lag setting in between two entered lag settings.

1. Press the button below the Lag Offset to display the “Manual Lag Adj” on the Conveyor Tracking Setup screen.
2. Operate the conveyor using a production low speed value.
3. Execute the program for the robot to pick parts, and fine tune the Lag Offset setting until the robot picks the part in the center.
4. Enter these values into the table in the Local Tags of the MLxConveyorSync_0 program.
   – LagAdjust[0].Speed = the low speed setting (mm/s)
   – LagAdjust[0].Lag = the lag setting to pick in the center (mm)

Fig. 2-22: Speed/Lag Offset Lookup Table



5. Increase the conveyor speed, and repeat *step 3* and *step 4* using multiple production speed settings.
   – The entries need to be in the order of increasing speed, where the last entry is the fastest speed.

NOTICE

Enter up to ten conveyor speeds. The routine will interpolate between these settings to determine the correct Lag Offset setting for any speed.

2.8 Using Simulation or Drives Feedback instead of an Encoder for Conveyor Position

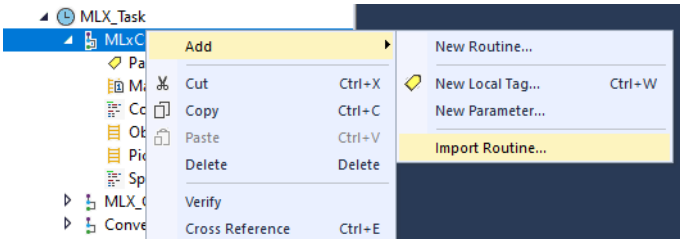
It is possible to use a simulated position generator or the conveyor servo drive feedback instead of the encoder and HSC counter card. The drive will need to be setup to send the conveyor position data to the PLC over Ethernet. Change the PLC ladder with the following steps.

NOTICE

This modified Main routine is for a single conveyor tracking robot. Similar changes can be added to the dual robot conveyor sync program.

1. Right-click on the MLxConveyorSync_0 program, press “Add”, and choose “Import Routine”.

Fig. 2-23: Importing new MLxConveyorSync Main routine for Simulation or Drives Feedback



2. Browse the software CD folder “Options - Simulation or Drives Position Feedback”, and select “Main_for_Sim_or_Drives_FB”.
   – This imports a modified Main routine.
3. Open the new Main routine of the MLxConveyorSync_0 program and observe the comments at the top to either setup a simulated position generator or the feedback from a servo drive.

2.9 Using Simulated Conveyor Position Feedback

There is a choice to setup a simulated conveyor position feedback. Use Rung 0 for the simulation setup. Set the following tags to create a simulated conveyor feedback at a defined speed.

– Rung 0: SetPeriod - enter the period setting of the MLX_Task (ms)

– Rung 0: SetSpeed - enter the desired speed (mm/s)

– Rung 0: EnableSimulation - turn on to enable the simulated conveyor position feedback.

– Rung 2: Sensor - toggle to simulate a part crossing the sensor.

2.10 Using Drive Position Feedback and a Part Sensor Input

Modify the following rungs to use the positional feedback sent from the servo drive and a part sensor input.

- Rung 1 - replace the “ConveyorPos” tag with the position data tag sent from the servo drive. The units of the position needs to be in mm.
- Rung 2 - replace the “Sensor” tag with the part detection tag that the sensor turns on.

3 Developing a Conveyor Tracking Application

The following sections will describe the basic methodology to program a conveyor tracking application.

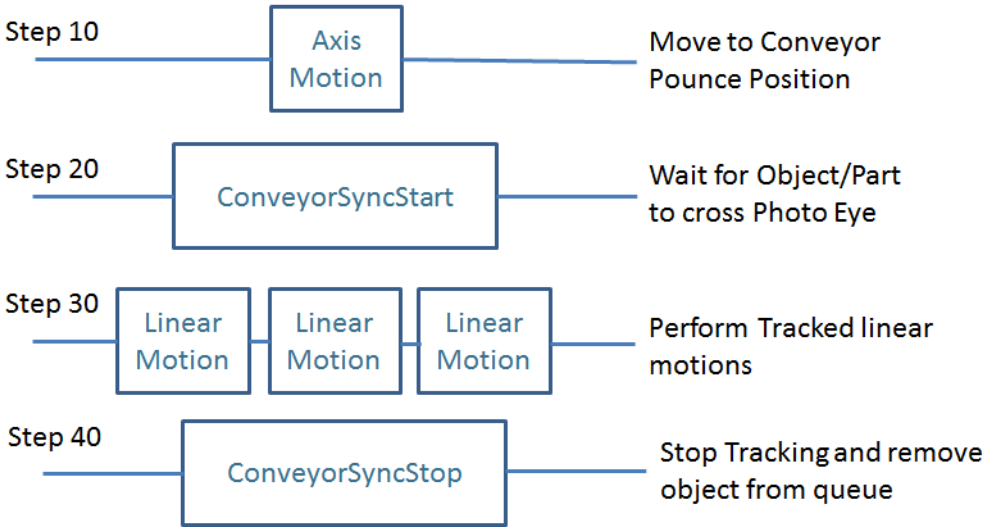
3.1 Program Structure Overview

Fig. 3-1 "Basic Program Structure for a Conveyor Tracking Application" shows the basic structure of a simple Conveyor Tracking program. On Step 10, an Axis Motion (e.g. MLxRobotMoveAxisAbsolute) is used to move the robot to its "pounce" position. This is the position where the robot will wait to start tracking objects. After this motion is complete, Step 20 will wait at an MLxRobotConvSyncStart command until a part crosses the photo eye and moves past the defined ConveyorStartPosition. After this happens, the program will move to Step 30 where the linear motions to the taught points relative to the part are executed while tracking the moving part. When these motions are complete, Step 40 will call an MlxRobotConvSyncStop command which will stop the tracking action and also remove the part from the queue.

NOTICE

- The points taught for the linear motion during conveyor tracking, need to be taught in world coordinate, not in the conveyor user frame.
- Only linear moves are supported when conveyor tracking is on. Any other type of motion will result in an error.

Fig. 3-1: Basic Program Structure for a Conveyor Tracking Application



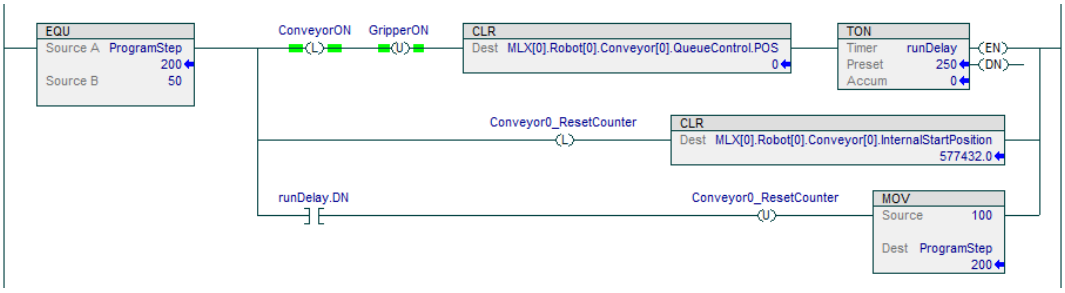
ConvSyncStopWithAxisMot or ConvSyncStopWithLinearMot can replace ConvSyncStop in Step 40 to allow smooth blending out of the conveyor tracking operations. Using these two commands reduce the cycle time.

3.1.1 Program Structure Details

This section will show an example of programming a PLC routine to track a part. Refer to the ConveyorTrackingExample PLC program that was imported in section 2. Follow along the programmed steps as they are being explained. In the Startup routine, the first steps of the application is to activate the Tool and the User Frame that was taught for the conveyor. The next step is to reset the conveyor state (*Fig. 3-2 "Conveyor Initialization Step"*). The following actions are performed to reset the conveyor:

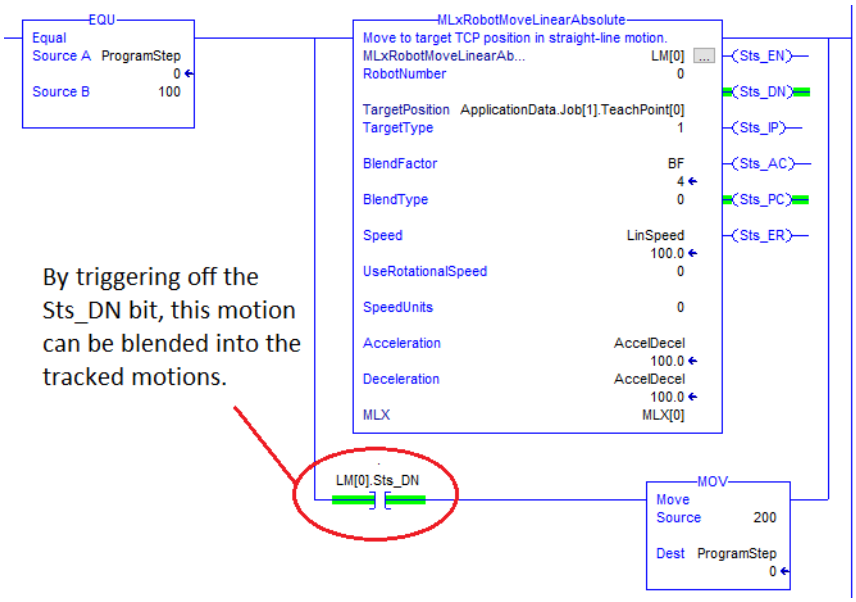
- Moving 0 into the robot's QueueControl.POS variable will act to flush the queue. Thus, any parts remaining in the queue will be removed.
- The Conveyor0_ResetCounter variable will simply reset the counter's current encoder count to 0.

Fig. 3-2: Conveyor Initialization Step



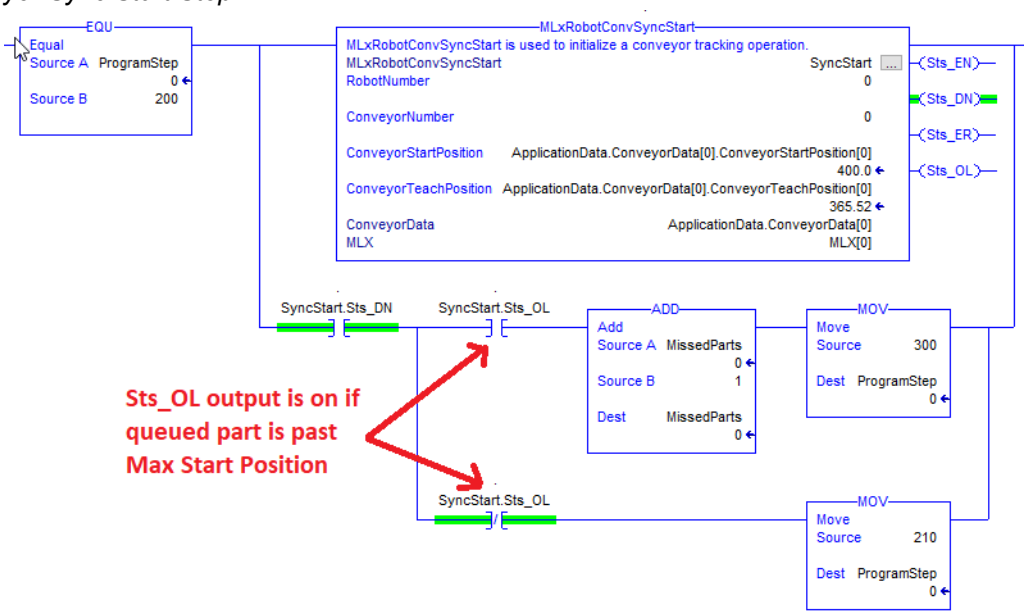
After the initialization in the Startup routine is complete, the program step is set to 100. In the Conveyor_Tracking routine, on step 100, a linear motion command will move the robot to the conveyor pounce position. Note that this step is incremented to 200 when the Sts_DN bit (instead of Sts_PC bit) of the motion instruction turns on. This means that the program will move to Step 200 (Conveyor Sync Start) as soon as the motion is queued rather than waiting for the motion to complete. Thus, if a part crosses the photo eye while this motion is taking place, it can blend into the tracked motions.

Fig. 3-3: Move to Conveyor Pounce Position



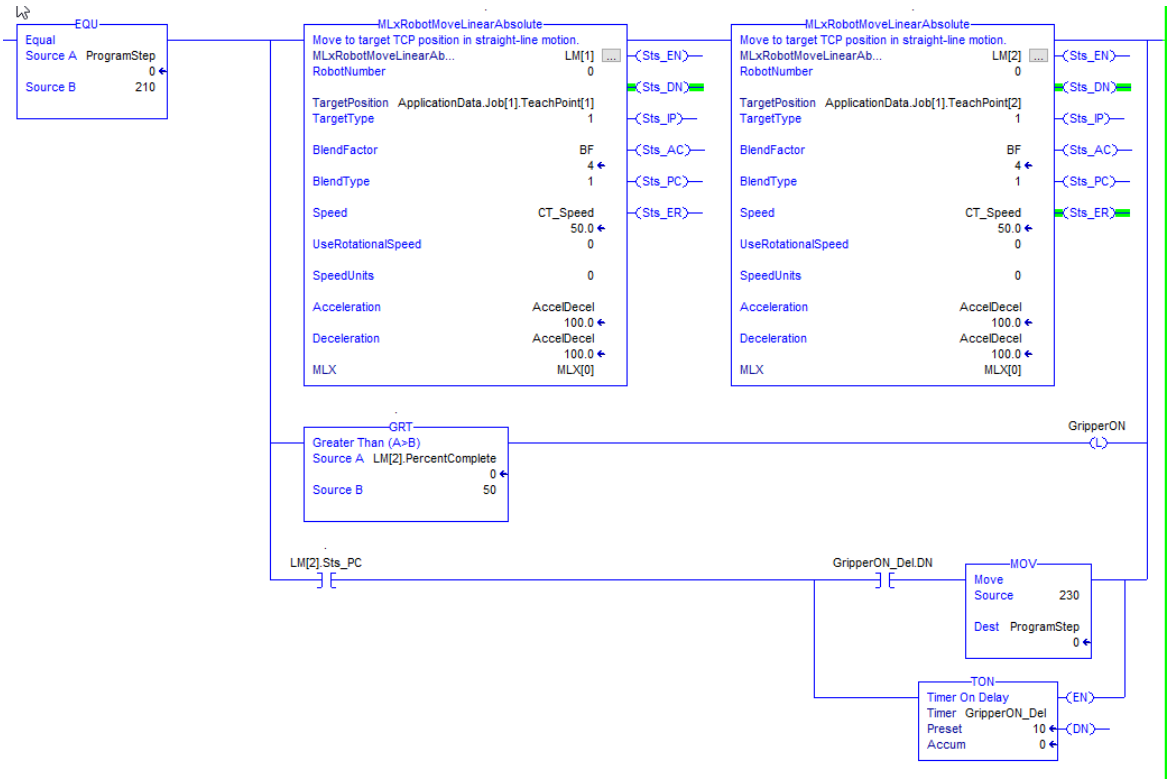
On Step 200, the Conveyor Sync Start instruction will wait for an object to pass the sensor. When a part is detected, the Sts_DN bit will turn on. This instruction also has a Sts_OL bit (overlimit) which will turn on if the object moves past its Max Start Position. If the Sts_OL bit turns on, the application will advance to the Conveyor Sync Stop rung (Step 300.) Otherwise, it will move to the Tracked Motions (Step 210.) The reason the program needs to execute the Sync Stop command for a missed part is to flush this part from the queue. (*Fig. 3-4*).

Fig. 3-4: Conveyor Sync Start Step



On Step 210, there are two linear motion commands that will move the gripper to the part, while conveyor tracking. The vacuum gripper turns on when the second motion is 50% completed.

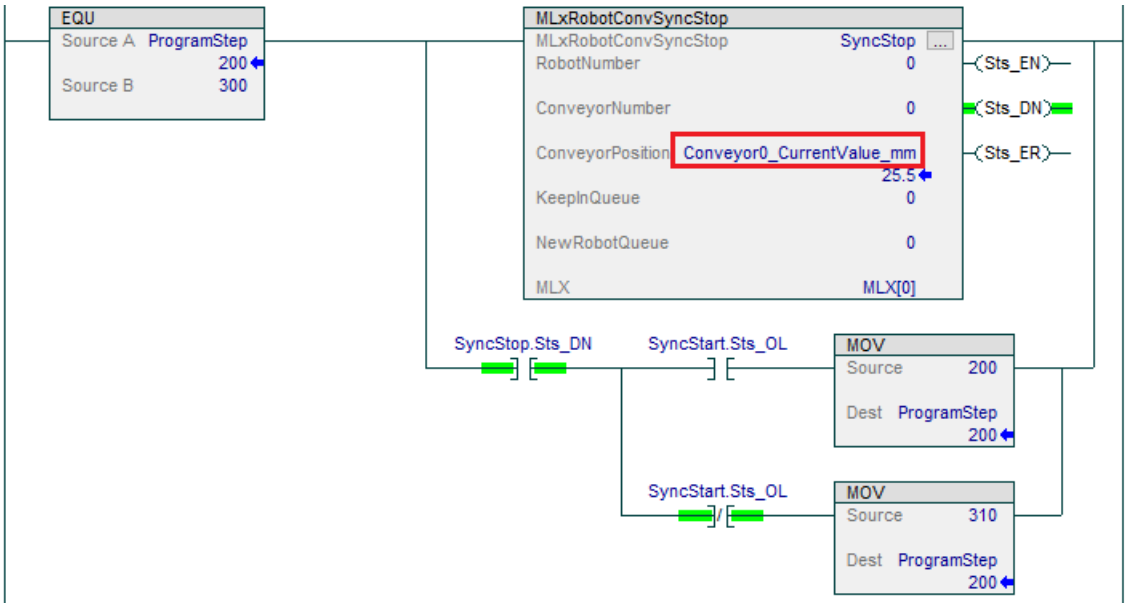
Fig. 3-5: Execute Tracked Motions Step



The robot will lift the part on Step 230, while still tracking (refer to program).

On Step 300, the Conveyor Sync Stop instruction will turn off conveyor tracking operations and update the internal object queue (based on the values of KeepInQueue and NewRobotQueue as detailed in *section n "ConveyorSyncStop AOIs" on page 4-2*. After the Sts_DN bit is set, the Sts_OL will be checked to determine if this part was out of reach. If it was, the program will return to the Conveyor Sync Start step (Step 200). If a part was actually picked, the program will advance to Step 310 to move to a placement location (refer to program).

Fig. 3-6: Conveyor Sync Stop Step

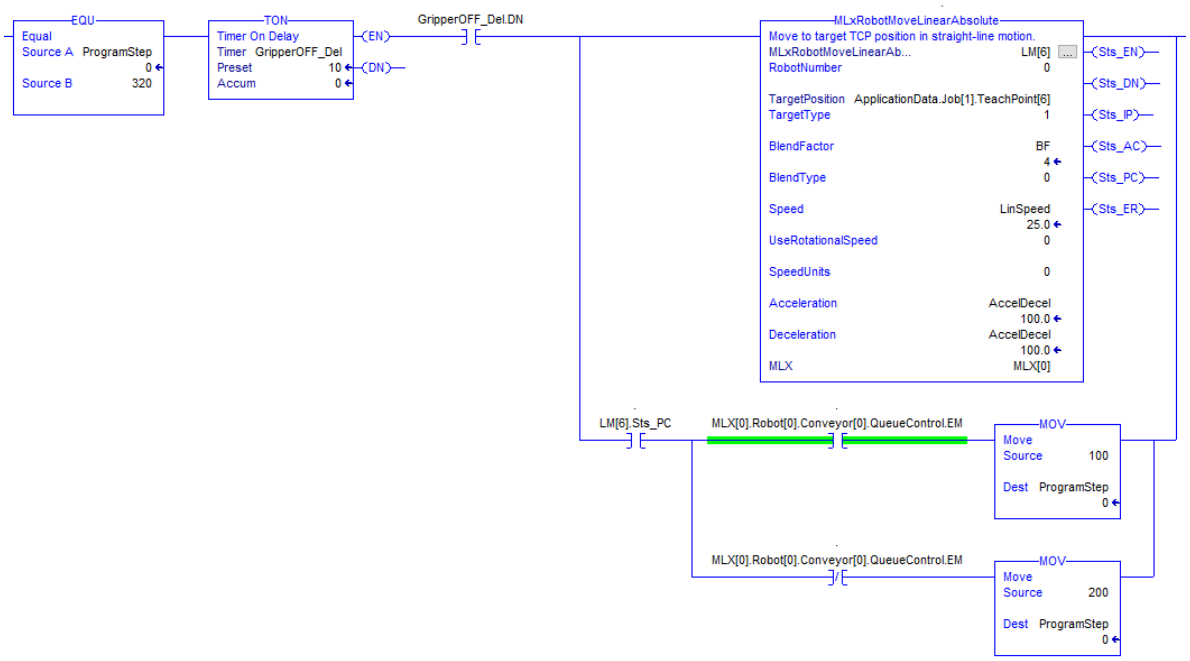


NOTICE

It is important that the "Conveyor0_CurrentValue_mm" is used for the ConveyorPosition, as shown in *Fig. 3-6*. This tag is in units of mm. The Conveyor0_CurrentValue tag is in units of encoder pulse counts and will not work.

After the part is placed, the program will advance to Step 320. There will be a small delay for the gripper release, and the robot will move up to a depart location. If the part queue is empty, the program will return to Step 100 to move back to the pounce position. However, if there is a new part already in the queue, the program will skip to Step 200 to immediately start tracking the part from this location.

Fig. 3-7: Depart from Part Placement



There are example rungs at the bottom of the ConveyorTrackingExample program that show how to use the MLxRobotConvSyncStopWithLinMot and MLxRobotConvSyncStopWithAxisMot commands.

3.2 Robot Number Assignments and Controlling Pick Order for Dual Robots

This section explains how to set the controller and robot numbers for the robots that are conveyor tracking.

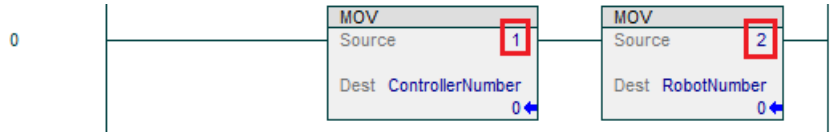
Examples show how the system can dynamically balance the picking order of two robots.

3.2.1 Single Robot Conveyor Tracking

The PickControl routine is modifiable in the MLxConveyorSync_0 program.

– Rung 0 - Modify the numbers for the ControllerNumber and RobotNumber

Fig. 3-8: Single Robot Controller and Robot Number Assignments



3.2.2 Two Robots on the Same Controller Conveyor Tracking

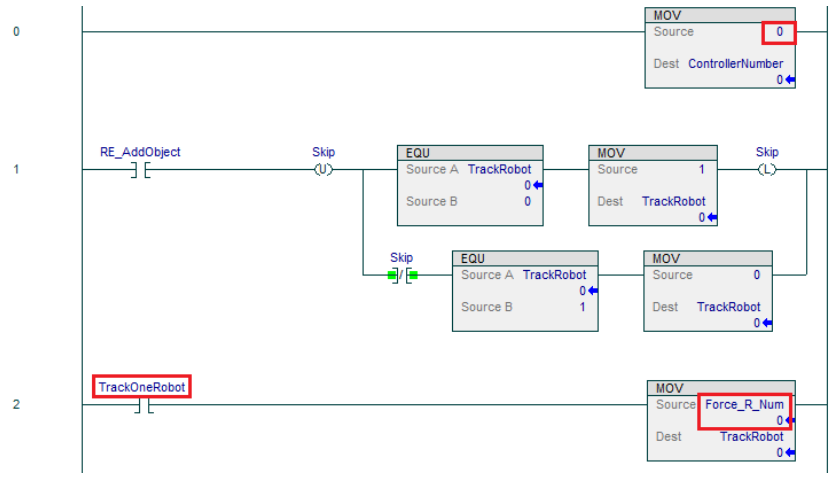
The PickControl routine is modifiable in the MLxConveyorSync_0_R0R1 program. This will alternate the tracking of two robots on the same controller. For this case, the robot numbers must be R0 and R1. If wanting to track only one robot, toggle on the “TrackOneRobot” tag.

– Rung 0: - Modify the number for the ControllerNumber

– Rung 2: - TrackOneRobot - Turn this on to force the tracking of one robot

– Rung 2: - Force_R_Num - Set this tag to force a certain robot number to track

Fig. 3-9: Two Robots on the Same Controller Picking Control

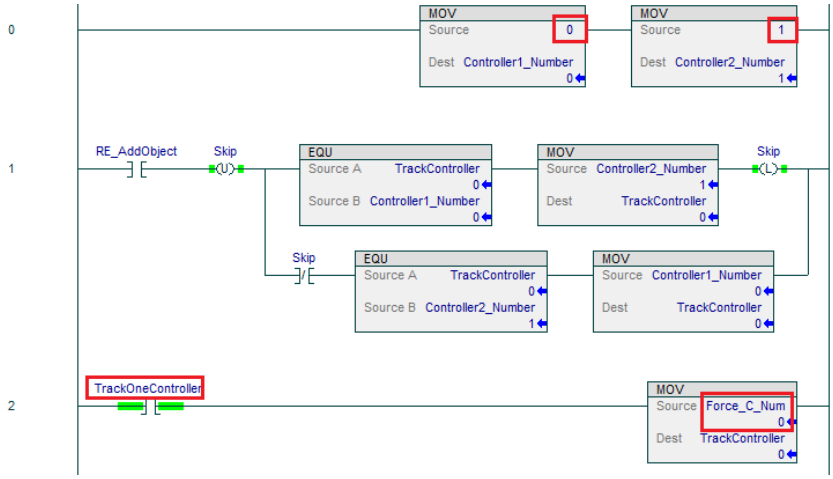


3.2.3 Two Robots on Different Controllers Conveyor Tracking

The PickControl routine is modifiable in the MLxConveyorSync_0_C0C1 program. This alternates the tracking of two robots on different controllers. For this case, the robot numbers on each controller must be R0. If wanting to track only one robot, toggle on the “TrackOneController” tag.

- Rung 0: - Modify the numbers for the Controller1_Number and Controller2_Number
- Rung 2: - TrackOneController - Turn this on to force the tracking on one robot
- Rung 2: - Force_C_Num - Set this tag to force robot 0 on a certain controller to track

Fig. 3-10: Two Robots on the Different Controllers Picking Control



4 Conveyor Tracking Commands (AOIs)

This section will describe each of the conveyor tracking AOIs. For additional information, refer to the Appendix A of the MLX300 SOFTWARE AND OPERATING USER'S MANUAL (180247-1CD).

4.1 Conveyor Tracking Instructions

The Robot Controller has four instructions related to Conveyor Tracking. There are two basic instructions to turn conveyor tracking sync on/off:

- **MLxRobotConvSyncStart**
  Starts conveyor tracking

- **MLxRobotConvSyncStop**
  Stops conveyor tracking

  There are also two instructions for blending motion within the sync commands. These commands are available to assist with increasing pick rates.

- **MLxRobotConvSyncStopWithLinearMot**
  While turning off conveyor tracking operation, blend motion into a linear path to the next point.

- **MLxRobotConvSyncStopWithAxisMot**
  While turning off conveyor tracking operation, blend motion into a joint path to the next point.

4.1.1 MLxRobotConvSyncStart Instruction

- **MLxRobotConvSyncStart**
  AOI is used to begin conveyor tracking operations. When called, this AOI will wait for an object to be added to the queue (this happens automatically when an object passes the photo eye) and then turn on the Sts_DN bit. *Each time a part crosses the photo eye, it is added to the queue.*

  Fig. 4-1: Conveyor Synchronization Start Instruction

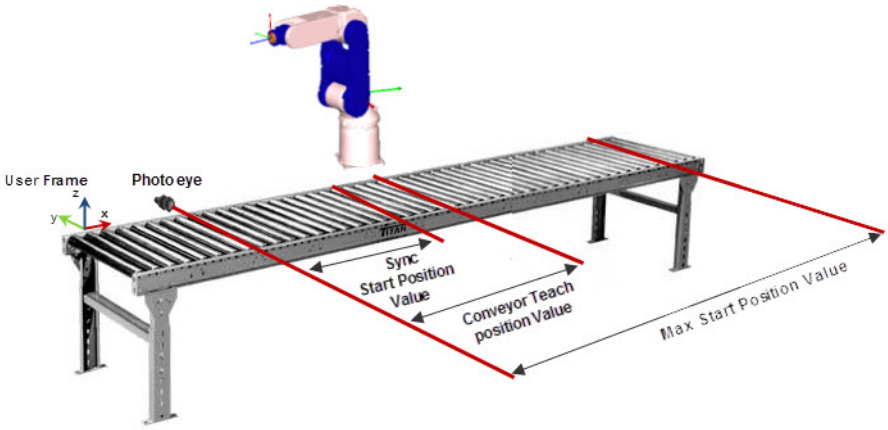
- **ConveyorStartPosition**
  This is the position along the conveyor that the robot will start to track queued objects.

- **ConveyorTeachPos**
  The ConveyorTeachPos is the position along the conveyor where the robot programmed points were taught with the conveyor stopped.

- **.Sts_DN**
  This output will be asserted once a part has crossed the ConveyorStartPos.

- **.Sts_OL**
  This output is asserted when a queued object has gone beyond the MaxStartPosition value.

Fig. 4-2: MLxRobotConvSyncStart Instruction Parameters



4.1.2 ConveyorSyncStop AOIs

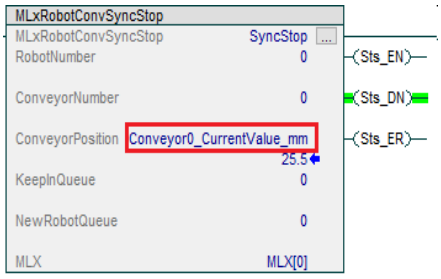
CAUTION

- Use “Conveyor0_CurrentValue_mm” for the conveyor position of the three ConveyorSyncStop AOIs.
  - Tag units are in mm.

Using the Conveyor0_CurrentValue tag will not work, because it’s units are encoder counts.

- **MLxRobotConvSyncStop Instruction**
  MLxRobotConvSyncStop turns off the conveyor synchronization and updates the internal object queue. The Conveyor0_CurrentValue for the correct conveyor must pass into the AOI. In addition, there are two parameters (KeepInQueue and NewRobotQueue) which are available to handle how the object queue updates. When setting KeepInQueue to 0, it flushes the object from the queue and tracking is not available. In this case, the value of NewRobotQueue does not matter. If KeepInQueue is set to 1, the value of NewRobotQueue moves the object to that robot's queue. If NewRobotQueue equals the current robot, the object is in the current robot's queue and is the first object in the queue when calling another ConvSyncStart. This is useful if an application requires stopping conveyor tracking operations temporarily while still tracking the same object.

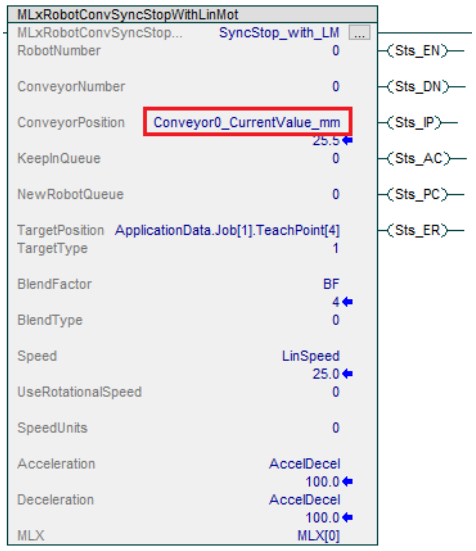
Fig. 4-3: Conveyor Synchronization Stop Instruction



■ MLxRobotConvSyncStopWithLinearMot Instruction

MLxRobotConvSyncStopWithLinearMot is basically a combination of a Linear Motion and a Conveyor Sync Stop. This instruction will turn off conveyor tracking operations and update the object queue like a normal Conveyor Sync Stop. However, instead of stopping the robot motion, it will instead blend directly into a linear motion to another point. This can help decrease cycle times in demanding applications. The rest of the parameters for the Linear Motion are identical to a normal motion instruction.

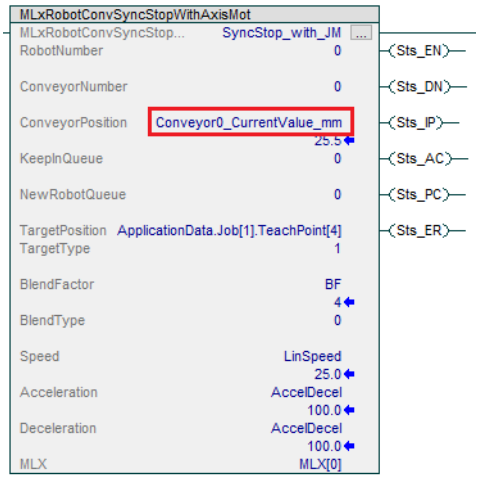
Fig. 4-4: Conveyor Synchronization Stop Followed by a Linear Move Instruction



■ MLxRobotConvSyncStopWithAxisMot Instruction

MLxRobotConvSyncStopWithAxisMot is a combination of an Axis Motion with a Conveyor Sync Stop. This instruction turns off conveyor tracking operations and updates the object queue like a normal Conveyor Stop. However, instead of stopping the robot motion, it will blend directly into an axis motion to another point. This can help decrease cycle times in demanding applications. The rest of the parameters for the Axis Motion are identical to a normal motion instruction.

Fig. 4-5: Conveyor Synchronization Stop Followed by an Axis Move Instruction



5 Debugging Pickup Position Errors

NOTICE

Use this section after developing the ladder program.

5.1 Grips at Different Locations on Part

If the part is gripped at different locations on the part, the conveyor resolution is not as accurate as it needs to be. This is typically observed when the part is picked from different locations on the conveyor.

To improve this resolution, slow the conveyor down, and add a long delay in the program after the move position above the part. During the playback, the robot will be tracking the conveyor with the gripper above the part. If the conveyor resolution tag is accurate, the robot should maintain its position relative to the moving part, while it is executing the long delay. If it is observed that the robot gripper is drifting forward or backward in respect to the part, modify the EncoderToMMConversion tag until the drift is eliminated.

5.2 Part is Gripped Consistently at the Wrong Location

If picking the part up anywhere on the conveyor with the gripper consistently at the wrong location on the part, there is a small execution delay that needs calibration. Typically, the response time of hardware I/O needs calibration. In most cases, the program will have the gripper pick up the part at a center location on the part.

There are two methods to achieve an accurate pick location on the part. (Refer to *section 2.6.4 “Teach Points and Setup Tracking Parameters” on page 2-16*)

- Manual Lag Adj mode - allows manually entering the Lag Offset value on the HMI to fine tune the pickup accuracy. This is a good method if the conveyor speed does not vary much during production.
- Auto Lag Adj mode - this enables a lookup table routine to set the correct Lag Offset based on the conveyor speed feedback. This is a requirement when the conveyor speed changes a lot during production.

There is an alternative method to calibrate a system to increase the picking accuracy when the conveyor speed various. YASKAWA can change MF1G004 Robot Controller parameter. This will be a process of testing low, medium and high speed settings for each MF1G004 test value until an optimal value can be determined.

5.3 Excessive Segment Alarm

When the motion command plus the conveyor speed exceeds the robot max speed, there will be an Excessive Segment alarm. To resolve this, lower the speed of the either the motion command or the conveyor. Sometime, increasing the “# of Points to Average” on the conveyor tracking screen can help eliminate this alarm.

MLX300
SOFTWARE OPTIONS
USERS MANUAL

FOR: CONVEYOR TRACKING

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Specifications are subject to change without notice
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